

James Song

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Education

University of Michigan

May 2026

B.S.E. Computer Science and Engineering, B.S. Mathematics

GPA: 3.94/4.00

- Coursework: Machine Learning Theory • GPU Programming • Generative Models • Foundations of LLMs • Linear Algebra • Operating Systems • Probability Theory • Discrete Stochastic Processes

Research Experience

Research Assistant, UofM Department of ECE – Ann Arbor, MI

Feb 2025 – Present

- Designed and trained NAMD, a multimodal latent diffusion model with a U-Net backbone conditioned on prior scans and MedGemma 4B clinical text embeddings via cross-attention for spatiotemporal medical image prediction, achieving 0.805 AUROC vs. 0.742 baseline, surpassing all prior methods. (*In submission to MICCAI 2026.*)
- Architected a custom VAE with a five-component loss (L1, KL, LPIPS, alignment, predictive representation) to structure the latent space by clinical attributes; explored conditioning strategies (cross-attention, DAFT, SFT, SPADE) across U-Net and DiT backbones.

Research Intern, CMU InfiniAI Lab – Pittsburgh, PA

May 2025 – Oct 2025

- Developed a black-box attack exposing Functionally Correct yet Vulnerable (FCV) patches in LLM code agents (SWE-Agent, OpenHands) evaluated in Docker-based sandboxes, achieving 18–63% attack success rates across 12 model-agent configurations on SWE-Bench. (*In submission to ACL 2026.*)

Research Assistant, UofM Department of CSE – Ann Arbor, MI

Oct 2024 – July 2025

- Investigating multi-choice reinforcement learning for LLM post-training as an alternative to DPO on Llama 3.2 1B/3B, Llama 3.1 8B, and Qwen 2.5 1.5B, using a reward model to score and weight multiple candidate responses, achieving a ~54% win rate against DPO-trained models on AlpacaEval.
- Implementing distributed training pipelines with HuggingFace Accelerate for LoRA post-training and inference.
- Built a synthetic preference data generation pipeline using vLLM for large-scale inference and benchmarked models with lm-evaluation-harness.

Research Assistant, UofM Transportation Institute – Ann Arbor, MI

May 2023 – Aug 2024

- Developed an automated mesh morphing pipeline using Adaptive Radial Basis Functions and Coherent Point Drift for T1–12 vertebrae, reducing maximum geometric error by 75% with sub-millimeter mean-squared error in MATLAB.

Projects

Linear Mode Connectivity (LMC) Experiment

June 2024 – July 2024

- Reimplemented GPT-2 (124M params) from scratch in PyTorch, including multi-head self-attention, layer normalization, and the full training loop with AdamW optimizer and learning rate scheduling.
- Replicated LMC experiments across LeNet, ResNet20, and decoder-only transformers based on Frankle et al. (2020), analyzing loss landscape geometry across vision and language architectures.

Real-Time Race Strategy Simulator (UM Solar Car Team)

Sept 2022 – April 2024

- Built a drive-by-second simulator in C++ modeling real-time energy, speed, and weather over 3,000 km solar car races, and a live cloud coverage radar integrating weather APIs with a Python Flask backend for race-day decision-making.

Skills

Programming Languages: Python, C++, C, MATLAB, Go

Tools and Frameworks: PyTorch, Accelerate, CUDA, NumPy, HuggingFace Diffusers, vLLM, transformers, wandb, Docker, Git